

```
mysql> use cg_mysql;
Database changed
mysql> /* select * from customers;
/*> */
```

```
mysql> select * from customers;
```

customer_id	customer_name	city	customer_type
1	Rahul	Delhi	Premium
2	Meena	Mumbai	Regular
3	Arjun	Bangalore	Premium
4	Kavya	Delhi	Regular
5	Suresh	Chennai	Premium
6	Anjali	Mumbai	Regular
7	Vikram	Delhi	Regular
8	Pooja	Bangalore	Premium
9	Devi	Chennai	Regular

```
9 rows in set (0.05 sec)
```

```
mysql> select * from products;
```

product_id	product_name	category	price
1	Laptop	Electronics	55000.00
2	Mobile	Electronics	25000.00
3	Chair	Furniture	7000.00
4	Table	Furniture	9000.00
5	Headphones	Electronics	3000.00
6	Bookshelf	Furniture	6000.00

```
6 rows in set (0.01 sec)
```

```
mysql> select * from orders;
```

order_id	customer_id	product_id	quantity	order_date	amount
1	1	1	1	2023-01-05	55000.00
2	2	2	1	2023-01-10	25000.00
3	3	3	2	2023-02-01	14000.00
4	1	5	2	2023-02-15	6000.00
5	4	4	1	2023-03-03	9000.00
6	5	2	2	2023-03-20	50000.00
7	6	4	3	2023-04-02	18000.00
8	3	5	1	2023-04-15	3000.00
9	7	3	1	2023-05-01	7000.00
10	8	1	1	2023-05-20	54000.00
11	2	3	2	2023-06-02	12000.00
12	5	4	1	2023-06-15	8500.00

12 rows in set (0.02 sec)

```
mysql> select product_name from products where price > all(select price from
products where category = 'furniture');
```

product_name
Laptop
Mobile

2 rows in set (0.04 sec)

```
mysql> select product_name from products where price < all(select price from
products where category = 'furniture');
```

product_name
Headphones

1 row in set (0.00 sec)

```
mysql> select order_id, amount from orders where amount > all(select amount from
orders where customer_id = (select customer_id from customers where customer_name =
'Meena'));
```

order_id	amount
1	55000.00
6	50000.00
10	54000.00

3 rows in set (0.05 sec)

```
mysql> -- nested subqueries
```

```
mysql> select customer_name from customers where customer_id in (select customer_id
from orders where product_id = (select product_id from products where price =
(select max(price) from products)));
```

customer_name
Rahul
Pooja

2 rows in set (0.05 sec)

```
mysql> select category from products where product_id = (select product_id from
orders where amount = (select max(amount) from orders));
```

category

```
| Electronics |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select city from customers where customer_id in (select customer_id from
orders where product_id = (select product_id from products where price = (select
min(price) from products)));
```

```
+-----+
| city |
+-----+
| Delhi |
| Bangalore |
+-----+
2 rows in set (0.00 sec)
```

```
mysql> select customer_name from customers c where (select count(*) from orders o
where o.customer_id = c.customer_id)>1;
```

```
+-----+
| customer_name |
+-----+
| Rahul |
| Meena |
| Arjun |
| Suresh |
+-----+
4 rows in set (0.00 sec)
```

```
mysql> select p1.product_name, p1.category, p1.price from products p1 where 1 =
(select count(*) from products p2 where p2.category = p1.category and p2.price =
p1.price);
```

```
+-----+-----+-----+
| product_name | category | price |
+-----+-----+-----+
| Laptop | Electronics | 55000.00 |
| Mobile | Electronics | 25000.00 |
| Chair | Furniture | 7000.00 |
| Table | Furniture | 9000.00 |
| Headphones | Electronics | 3000.00 |
| Bookshelf | Furniture | 6000.00 |
+-----+-----+-----+
6 rows in set (0.04 sec)
```

```
mysql> select p1.product_name, p1.category, p1.price from products p1 where 1 =
(select count(*) from products p2 where p2.category = p1.category and p2.price <
p1.price);
```

```
+-----+-----+-----+
| product_name | category | price |
+-----+-----+-----+
| Mobile | Electronics | 25000.00 |
| Chair | Furniture | 7000.00 |
+-----+-----+-----+
```

```
+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> select customer_name from customers c where exists(select 1 from orders o
where o.customer_id = c.customer_id);
```

```
+-----+
| customer_name |
+-----+
| Rahul         |
| Meena         |
| Arjun         |
| Kavya         |
| Suresh        |
| Anjali        |
| Vikram        |
| Pooja         |
+-----+
```

```
8 rows in set (0.00 sec)
```

```
mysql> select customer_name from customers c where not exists(select 1 from orders
o where o.customer_id = c.customer_id);
```

```
+-----+
| customer_name |
+-----+
| Devi          |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select * from employees;
```

```
+-----+-----+-----+-----+
| emp_id | name   | department_id | salary |
+-----+-----+-----+-----+
| 101    | Arjun  | 1             | 50000.00 |
| 102    | Bhavana | 2             | 65000.00 |
| 103    | Charan | 2             | 72000.00 |
| 104    | Divya  | 3             | 45000.00 |
| 105    | Eswar  | NULL          | 40000.00 |
| 106    | Farhan | 4             | 55000.00 |
| 107    | Gowri  | 2             | 62000.00 |
| 108    | Harish | 5             | 48000.00 |
| 109    | Ishita | NULL          | 53000.00 |
| 110    | Jagan  | 3             | 60000.00 |
+-----+-----+-----+-----+
```

```
10 rows in set (0.02 sec)
```

```
mysql> -- second min, third max and third min
```

```
mysql> select min(salary) from employees where salary > (select min(salary) from
employees);
```

```
+-----+
| min(salary) |
+-----+
```

```
+-----+
| 45000.00 |
+-----+
```

1 row in set (0.00 sec)

```
mysql> select min(salary) as third_min from employees where salary > (select
min(salary) as second_min from employees where salary > (select min(salary) from
employees));
```

```
+-----+
| third_min |
+-----+
| 48000.00 |
+-----+
```

1 row in set (0.01 sec)

```
mysql> select max(salary) as third_max from employees where salary < (select
max(salary) as second_max from employees where salary < (select max(salary) from
employees));
```

```
+-----+
| third_max |
+-----+
| 62000.00 |
+-----+
```

1 row in set (0.00 sec)

```
mysql> select d.department_id, d.department_name, min(e.salary) as
second_min_salary from employees e join departments d on e.department_id =
d.department_id where e.salary > (select min(salary) from employees e2 where
e2.department_id = e.department_id) group by d.department_id;
```

```
+-----+-----+-----+
| department_id | department_name | second_min_salary |
+-----+-----+-----+
| 2 | Engineering | 65000.00 |
| 3 | Marketing | 60000.00 |
+-----+-----+-----+
```

2 rows in set (0.06 sec)

```
mysql> select d.department_id, d.department_name, max(e.salary) as
second_max_salary from employees e join departments d on e.department_id =
d.department_id where e.salary < (select max(salary) from employees e2 where
e2.department_id = e.department_id) group by d.department_id;
```

```
+-----+-----+-----+
| department_id | department_name | second_max_salary |
+-----+-----+-----+
| 2 | Engineering | 65000.00 |
| 3 | Marketing | 45000.00 |
+-----+-----+-----+
```

2 rows in set (0.00 sec)

mysql>

```

mysql> /* Views
    /*> A view is a virtual table created from the result of select query. It
does'nt normally store the actual data separately instead it stores the SQL Query
and whenever you access the view,
    /*> MySQL retrieves the data from the underlying tables.
    /*>
    /*> Why do we need views?
    /*> 1. simplify complex queries.
    /*> 2. reuse query logic.
    /*> 3. data security.
    /*> 4. simplified reporting.
    /*>
    /*> Advantages of views
    /*> 1. simplicity
    /*> 2. security
    /*> 3. reusability
    /*> 4. consistancy
    /*> 5. maintainability
    /*> 6. no extra storage cost
    /*>
    /*> difference between table and view
    /*> - table physically stores data -- view is virtual table based on query
    /*> - can store records directly -- usually doesn't store separate records
    /*> - create using create table -- create using create view
    /*> - contains all the data -- can show selected rows/columns
    /*> -used for permanent data storage -- mainly used for abstract, security, and
reusable queries.
    /*>
    /*> syntax: create view view_name as select column_name from table_name where
condition;
    /*>
    /*> alter view: alter view view_name as select new_columns from new_tables where
new_conditions;
    /*>
    /*> drop view: drop view view_name;
    /*> */
mysql>
mysql> drop table employees;
Query OK, 0 rows affected (0.06 sec)

mysql> drop table departments;
Query OK, 0 rows affected (0.08 sec)

mysql> CREATE TABLE departments (
    -> dept_id INT PRIMARY KEY AUTO_INCREMENT,
    -> dept_name VARCHAR(50),
    -> location VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.04 sec)

```

```
mysql>
mysql> CREATE TABLE employees (
  -> emp_id INT PRIMARY KEY AUTO_INCREMENT,
  -> emp_name VARCHAR(100),
  -> dept_id INT,
  -> designation VARCHAR(50),
  -> salary DECIMAL(10,2),
  -> join_date DATE,
  -> FOREIGN KEY (dept_id) REFERENCES departments(dept_id)
  -> );
```

Query OK, 0 rows affected (0.07 sec)

```
mysql>
mysql> INSERT INTO departments (dept_name, location) VALUES
  -> ('IT','Hyderabad'), ('HR','Bangalore'), ('Finance','Chennai'),
  -> ('Sales','Mumbai');
```

Query OK, 4 rows affected (0.01 sec)

Records: 4 Duplicates: 0 Warnings: 0

```
mysql>
mysql> INSERT INTO employees (emp_name, dept_id, designation, salary, join_date)
VALUES
  -> ('Rahul',1,'Software Engineer',55000,'2021-06-15'),
  -> ('Anita',2,'HR Executive',38000,'2020-03-21'),
  -> ('Suresh',4,'Sales Manager',52000,'2019-08-10'),
  -> ('Priya',1,'Team Lead',65000,'2022-01-05'),
  -> ('Kiran',3,'Finance Analyst',48000,'2018-12-11'),
  -> ('Deepika',2,'HR Manager',60000,'2019-10-19')
  -> ;
```

Query OK, 6 rows affected (0.01 sec)

Records: 6 Duplicates: 0 Warnings: 0

```
mysql> select * from employees;
```

emp_id	emp_name	dept_id	designation	salary	join_date
1	Rahul	1	Software Engineer	55000.00	2021-06-15
2	Anita	2	HR Executive	38000.00	2020-03-21
3	Suresh	4	Sales Manager	52000.00	2019-08-10
4	Priya	1	Team Lead	65000.00	2022-01-05
5	Kiran	3	Finance Analyst	48000.00	2018-12-11
6	Deepika	2	HR Manager	60000.00	2019-10-19

6 rows in set (0.00 sec)

```
mysql> select * from departments;
```

dept_id	dept_name	location
1	IT	Hyderabad

2	HR	Bangalore
3	Finance	Chennai
4	Sales	Mumbai

4 rows in set (0.00 sec)

mysql> create view emp_dept_name as

-> select e.emp_id, e.emp_name, e.designation, d.dept_name, d.location from employees e join departments d on e.dept_id = d.dept_id;

Query OK, 0 rows affected (0.01 sec)

mysql> show full tables where table_type = 'view';

Empty set (0.04 sec)

mysql> show full tables where table_type = 'VIEW';

Tables_in_cg_mysql	Table_type
emp_dept_name	VIEW

1 row in set (0.05 sec)

mysql> select * from emp_dept_name;

emp_id	emp_name	designation	dept_name	location
1	Rahul	Software Engineer	IT	Hyderabad
4	Priya	Team Lead	IT	Hyderabad
2	Anita	HR Executive	HR	Bangalore
6	Deepika	HR Manager	HR	Bangalore
5	Kiran	Finance Analyst	Finance	Chennai
3	Suresh	Sales Manager	Sales	Mumbai

6 rows in set (0.00 sec)

mysql> create view dept_summary as

-> select d.dept_name, count(e.emp_id) as total_employees, sum(e.salary) as total_salary, avg(e.salary) as average_salary from employees e

-> join

-> departments d on e.dept_id = d.dept_id group by d.dept_id;

Query OK, 0 rows affected (0.04 sec)

mysql> select * from dept_summary;

dept_name	total_employees	total_salary	average_salary
IT	2	120000.00	60000.000000
HR	2	98000.00	49000.000000
Finance	1	48000.00	48000.000000
Sales	1	52000.00	52000.000000

```
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> create view emp_annual_salary as
-> select emp_name, salary as monthly_salary, salary*12 as annual_salary
-> from employees;
```

Query OK, 0 rows affected (0.07 sec)

```
mysql> select * from emp_annual_salary;
```

```
+-----+-----+-----+
| emp_name | monthly_salary | annual_salary |
+-----+-----+-----+
| Rahul    | 55000.00      | 660000.00    |
| Anita    | 38000.00      | 456000.00    |
| Suresh   | 52000.00      | 624000.00    |
| Priya    | 65000.00      | 780000.00    |
| Kiran    | 48000.00      | 576000.00    |
| Deepika  | 60000.00      | 720000.00    |
+-----+-----+-----+
```

6 rows in set (0.04 sec)

```
mysql> create view highest_emp_summary as
-> select dept_name, average_salary from dept_summary where average_salary >
50000;
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> select * from highest_emp_summary;
```

```
+-----+-----+
| dept_name | average_salary |
+-----+-----+
| IT        | 60000.000000   |
| Sales     | 52000.000000   |
+-----+-----+
```

2 rows in set (0.00 sec)

```
mysql> alter view emp_dept_name as select e.emp_id, e.emp_name, d.dept_name from
employees e join departments d on e.dept_id = d.dept_id;
```

Query OK, 0 rows affected (0.06 sec)

```
mysql> select * from emp_dept_name;
```

```
+-----+-----+-----+
| emp_id | emp_name | dept_name |
+-----+-----+-----+
| 1      | Rahul    | IT        |
| 4      | Priya    | IT        |
| 2      | Anita    | HR        |
| 6      | Deepika  | HR        |
| 5      | Kiran    | Finance   |
| 3      | Suresh   | Sales     |
+-----+-----+-----+
```

6 rows in set (0.00 sec)

```
mysql> drop view highest_emp_summary;  
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> show full tables where table_type = 'VIEW';
```

Tables_in_cg_mysql	Table_type
dept_summary	VIEW
emp_annual_salary	VIEW
emp_dept_name	VIEW

3 rows in set (0.00 sec)

```
mysql> notee
```